EMMANUEL BURUVURU

Software Engineer | AI, Data & Cloud

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience designing and delivering full-stack applications, backend APIs, data pipelines and AI-enabled platforms across clinical research, cybersecurity and enterprise environments. Proficient in Python, Java and TypeScript with Django, Spring Boot, Node.js and React Native, deployed on Oracle Cloud, AWS and Azure using Docker and CI/CD. Hands-on with Retrieval-Augmented Generation (RAG), vector embeddings, LLM tool integration via Model Context Protocol (MCP) and ETL workflows. Builds secure, scalable systems that measurably improve operational efficiency and data accuracy. Oracle Cloud certified; pursuing an MSc in Applied AI and Data Science.

TECHNICAL SKILLS

Languages	Python, Java, JavaScript, TypeScript, SQL
AI & Data	RAG, Vector Embeddings, LLM Integration, Model Context Protocol (MCP), Machine Learning, ETL Pipelines, Data Engineering
Backend	Django, Django REST Framework, Spring Boot, Node.js, REST APIs, WebSockets, Webhooks, System Integration
Frontend & Mobile	React, React Native, Flutter, HTML, CSS
Databases	PostgreSQL, MySQL, MongoDB, Supabase, Query Optimization, Data Modeling
Cloud & DevOps	Oracle Cloud Infrastructure (OCI), AWS, Microsoft Azure, Docker, GitHub Actions, CI/CD
Security	JWT Authentication, RBAC, Secure API Design, Microsoft Entra ID, Microsoft Intune, Check Point, Rapid7, Vulnerability Assessment

PROFESSIONAL EXPERIENCE

Software Engineer — Africa Clinical Research Network (ACRN Health) Jul 2025 – Present

- Engineered AI-enabled operational applications processing 5,000+ records per day, improving workflow efficiency by 25%.
- Owned integrations across 10+ enterprise and clinical systems (CTMS, Oracle Clinical One EDC, RealTime eSource, eConsent, eDocs), streamlining workflows and reducing manual processing by 40%.
- Designed ETL pipelines and data workflows that improved data accuracy and processing speed by 30%.
- Built a fleet dispatch platform end to end — Django backend, Flutter mobile apps and GPS telematics integration — with an MCP integration enabling LLM-assisted dispatch operations.
- Led data migration mapping from RealTime eSource to Oracle Clinical One EDC and contributed to the organisation's Oracle Cloud (OCI) migration programme.

Software Developer / Associate Cybersecurity Engineer — Ignite Africa Jun 2024 – Jun 2025

- Developed automated reporting solutions and conducted security assessments for 4 client companies, remediating 50+ critical vulnerabilities before deployment.
- Secured 1,000+ endpoints using Check Point, Rapid7 and Microsoft Intune, reducing security incidents by 30–40%.
- Built Microsoft Power Apps solutions to support business continuity processes.

Software Engineer (Part-Time) — Accenture Feb 2023 – Nov 2023

- Built full-stack application features using Spring Boot, Node.js, MongoDB and modern frontend frameworks.
- Developed and deployed a secure internal learning platform, improving reliability, usability and deployment workflows.

EDUCATION

MSc Applied AI and Data Science (part-time) — Middlesex University Mauritius 2026 – Expected 2028

BSc Information Technology — Curtin University 2024

CERTIFICATIONS

- Oracle Cloud Infrastructure Certified Architect Associate
- Oracle Cloud Infrastructure AI Foundations Associate
- Oracle Cloud Infrastructure Foundations Associate

Verify: emmanuel.availafrica.com/certifications

LANGUAGES

English (Fluent) | French (Intermediate) | German (Intermediate)